

An investigation into Outdated Cybersecurity Practices in Healthcare institutions and Their Contribution to Ransomware Vulnerabilities

by

Student name: **Tshiamo Maise**

Student number: **602178**

Module: **Research Methods 381**

Course code: **RSH381**

Assignment number: **2**

Lecturer's name: **Dr Alfred Mazorodze**

Date of submission: **31/03/2026**

Table of Contents

Significance of the study.....	4
1.1. Introduction	5
1.2. Overview of ransomware attacks in the healthcare sector	5
1.3. Cybersecurity practices and staff awareness in healthcare institutions.....	7
1.4. Technical vulnerabilities that expose healthcare organisations to ransomware.....	8
1.5. The role of backup strategies in ransomware recovery	9
1.6. Cost effective cybersecurity measures for healthcare organisations	9
1.7. Gaps identified in the literature	10
1.1. Research Approach and Design.....	10
1.2. Target Population and Data Sources	11
1.3. Data Collection Method	11
1.4. Data Analysis Strategy	12
1.5. Validity, Reliability and Feasibility	12
1.6. Ethical Considerations.....	12
1.1. Type of results expected.....	13
1.2. How findings will address the research questions.....	14
1.3. Potential implications	14
References.....	17

Research Problem

Ransomware attacks have become a growing threat to the healthcare sector, where hackers deploy a malicious software that encrypts sensitive data and demand payment for its restoration (Mohammed, et al., 2025). The limited prioritisation of cybersecurity within healthcare institutions has left organisations unable to identify and mitigate cyber threats, thereby increasing their exposure to attacks and data breaches (Swasey, 2020). These attacks carry significant consequences for patients, as they expose patients' personal health information and interrupt the delivery of essential medical services, posing direct risks to patient safety and health outcomes (McGlave, et al., 2023). Furthermore, organisations face significant financial losses due to legal action that follows such incidents (Malenfant, 2021). This study aims to investigate how outdated cybersecurity practices contribute to ransomware vulnerabilities in healthcare institutions, with a focus on technical weaknesses and poor backup strategies, while identifying affordable measures to protect patients and reduce financial harm.

Research Questions

1. Why do current cybersecurity practices in healthcare institutions fail to prevent ransomware attacks effectively?
2. In what ways can the effectiveness of backup strategies in reducing the impact of ransomware on healthcare organisations be evaluated and what are the key success factors?
3. Which cybersecurity measures are the most cost-effective for healthcare organisations in preventing ransomware attacks?

Research Objectives

1. To identify and analyse the most critical technical vulnerabilities that expose healthcare organisations to ransomware attacks.
2. To assess the effectiveness of backup strategies in healthcare ransomware incidents in terms of the recovery time and the percentage of lost data.

3. To analyse and prioritise cybersecurity measures that healthcare organisations can implement to reduce ransomware risks, considering factors such as cost, feasibility and impact.

Significance of the study

Ransomware is not simply an IT problem, it is a patient safety problem, a financial problem and increasingly a national security problem (Technology, 2021). It is against this broader context that the significance of this study should be understood. Theoretically, this study contributes to a growing but still incomplete body of research. Much of the existing research examines ransomware threats in developed healthcare systems, where resources, regulatory frameworks and cybersecurity infrastructures are considerably more advanced than in developing countries (Neprash, et al., 2022). Studies conducted in the United Kingdom and United States dominates the existing literature, leaving a significant gap in knowledge regarding how these threats occur in resource-constrained settings (Swasey, 2020). By incorporating South African evidence alongside international research, this study offers a more contextually diverse perspective on ransomware vulnerabilities and contributes to a more academic understanding of the problem (Chuma, 2025). The study further extends this perspective by incorporating evidence on cybersecurity awareness gaps among healthcare staff in the South African context (Singh & Singh, 2022).

The practical contribution of this study is equally important. Evidence suggests that even when healthcare organisations recognise the threat of ransomware, they often lack the specific and affordable guidance on how to strengthen their defences (Swasey, 2020). Many existing cybersecurity frameworks are either too complex or too costly for under-resourced healthcare institutions to implement effectively (Mahmoud, 2024). This study directly responds to this challenge by identifying the most critical vulnerabilities, evaluating effective backup and recovery strategies and proposing a prioritised set of cost-effective security measures tailored to the financial realities of healthcare institutions (Patel, et al., 2024). The assessment of backup effectiveness through measurable outcomes such as recovery time and data loss, address an important gap in literature given that evidence shows only a small proportion of healthcare organisations are currently able to successfully restore data following a ransomware attack

(Neprash, et al., 2022). In doing so, the study contributes not only to academic knowledge but also to the work of healthcare managers, cybersecurity professionals and policy makers who are responsible for maintaining secure systems and protecting patient safety (Pavlova & Albert, 2025).

Literature Review

1.1. INTRODUCTION

The healthcare sector is experiencing a rapid digital transformation with many organisations adopting electronic health records (EHRs), telemedicine, Internet of Things(IoT) devices and cloud-based solutions to modernise service delivery and improve patient care (Shiraz, 2024). While these technological advances provide operational benefits, they have equally introduced significant cybersecurity challenges, placing healthcare institutions at considerable risk systems (Aldosari, 2025). Even where institutions have implemented security measures such as multi-factor authentication stricter user access controls, the rapid evolution of technology continues to introduce new vulnerabilities which cybercriminals exploit (Jones, et al., 2022). Given that the healthcare organisations routinely handle highly sensitive patient and financial data, this sector becomes lucrative targets for cybercriminals seeking unauthorized access to protected systems (Jones, et al., 2022). Among the various cyber threats facing healthcare, ransomware has proven to be one of the most damaging, capable to of jeopardising patient safety and causing permanent loss or compromise of essential medical information (Aldosari, 2025). According to Hundal et al. (2025), approximately 70% of digital transformation initiatives in healthcare sectors fail to meet their intended objectives due to organisational challenges including complex compliance requirements, cultural resistance limited IT workforce skills and difficulties in maintaining seamless data interoperability. These vulnerabilities leave critical security gaps unaddressed, making organisations more susceptible to advanced cyber threats.

1.2. OVERVIEW OF RANSOMWARE ATTACKS IN THE HEALTHCARE SECTOR

Ransomware is a form of malicious software that employs cryptographic techniques to encrypt a victim's files and system resources, rendering them inaccessible until a ransom, typically in demanded in cryptocurrency, is paid to the attacker (Alshaikh, et al., 2020). Unlike the traditional

malware, the impact of a ransomware attack is largely irreversible even after the malware has been removed, making recovery particularly destructive and difficult to recover without the attacker's cooperation (Alshaikh, et al., 2020). Ransomware commonly enters through a system through multiple vectors including phishing emails, malicious advertisements, social media links and Ransomware-as-a-Service platforms available on the dark web (Kapoor, et al., 2021). Once inside a system, the malware follows a structured attack process consisting of infection, encryption, demand and result, with the attackers using cryptocurrency payment addresses to maintain anonymity and avoid detection by law enforcement (Kapoor, et al., 2021). The financial impact of ransomware is substantial, with global losses projected to reach \$20 billion in 2021, a significant increase from \$325 million in 2015 (Alshaikh, et al., 2020).

While Alshaikh, Ramadan and Hefny (2020) focus primarily on the technical mechanics of ransomware, Kapoor et al. (2021) extends this analysis by emphasising on the role of cryptocurrencies in enabling large untraceable payments, arguing that the emergence of digital currencies has been a key factor driving the rapid growth of ransomware attacks. This distinction is important because it highlights that ransomware is not solely a technical issue, but one that is shaped by broader financial and regulatory dynamics. The Institute for Security and Technology (2021) supports this view, arguing that the ransomware cannot be solved through technical solutions alone and requires coordinated international efforts involving law enforcement, government and the private sector. The healthcare sector has emerged as the most heavily targeted industry, accounting for nearly 25% of all data breaches in 2020 with an average of breach cost exceeding \$7 million (Sumner & Keenan, 2022). Evidence from Neprash et al. (2022) documents 374 ransomware attacks on United States healthcare organisations between 2016 and 2021, exposing nearly 42 million patients' personal health information, with annual attacks doubling more than 43 to 91 over this period. Critically, these statistics are likely underestimates due to significant underreporting, highlighting a gap in current monitoring mechanisms that undermines prevention efforts (Neprash, et al., 2022).

1.3. CYBERSECURITY PRACTICES AND STAFF AWARENESS IN HEALTHCARE INSTITUTIONS

A consistent theme across the literature is that cybersecurity in healthcare has historically lagged behind other sectors due to underinvestment and insufficient staff awareness. Swasey (2020) notes that healthcare providers spent approximately 5% of their IT budgets on cybersecurity in 2018, considerably lower than other industry averages and that 39% of healthcare sector IT teams consisted of fewer than 10 people. This underinvestment leaves critical gaps in organisational defences, which cybercriminals can readily exploit. Kioskli et al (2021) support this view arguing that the main challenges contributing to ineffective cybersecurity management in healthcare include inadequate top management commitment, lack of formal security training programmes and the absence of transparent accountability mechanisms within the healthcare sector.

However, while both Swasey (2020) and Kioskli et al (2021) agree on the problem of underinvestment, they differ in their emphasis on solutions. Swasey (2020) argues primarily for increased budget allocation and infrastructure upgrades, whereas Kioskli et al (2021) adopts a broader sociotechnical perspective, arguing that cybersecurity risks extend beyond financial limitations and are deeply influenced by human behaviour and organisational dynamics. This distinction is significant because it suggests that interesting budgets without addressing the human element of cybersecurity.

The human element is a critical and often underestimated factor in healthcare cybersecurity. Singh and Singh (2022) investigated the cybersecurity knowledge and practice of nurses in private hospitals in KwaZulu-Natal, South Africa, finding significant misunderstanding and ignorance of basic cybersecurity concepts. The study revealed that only 21.7% of participants were aware of what ransomware is and fewer than half understood phishing, despite these being among the most common cyber threats targeting healthcare workers. Furthermore, approximately 47.2% of healthcare staff surveyed had never received any form of cybersecurity training and many tended to overestimate their own cybersecurity skills, creating a false sense of security that increases organisational vulnerability (Singh & Singh, 2022).

These findings are consistent with those of Kioskli et al (2021), who found that employee behaviour and negligence represent the most commonly exploited vulnerability in healthcare cybersecurity, with 46% of healthcare data breaches in 2017 attributed to employee behaviour such as clicking on phishing links. The South African context of (Singh & Singh, 2022) is particularly relevant to this study as it demonstrates that the cybersecurity awareness identified in international literature is evident within the local context reinforcing the need for intervention in the South African healthcare environment.

1.4. TECHNICAL VULNERABILITIES THAT EXPOSE HEALTHCARE ORGANISATIONS TO RANSOMWARE

Beyond human factors, the technical infrastructures of healthcare organisations present significant vulnerabilities that facilitate ransomware attacks. Patel et al. (2024) identifies several common vulnerabilities targeted by ransomware, including outdated operating systems, unpatched software, weak passwords, improper remote desktop service configurations and lack of multi-factor authentication. These technical weaknesses provide ransomware attackers with multiple entry points into healthcare systems, and the absence of adequate controls means that a single successful exploit can spread across an entire network.

The challenge of outdated systems is especially serious in the South African healthcare sector. Chuma (2025) found that 73% of public hospitals and clinics continue to use outdated systems such as Windows XP and Windows 7, which are no longer supported by security updates. The study also reported that widespread such as unpatched software, system misconfigurations, weak encryption and poor authentication mechanisms. While Patel et al. (2024) suggests that modern technologies like artificial intelligence can help detect and prevent ransomware, Chuma (2025) highlights that financial constraints, shortage of skilled IT staff and resistance to change makes it difficult to implement these solutions in resource-limited environments. Al-Qarni (2023) and Kioskli et al. (2021) further support these findings by noting that poor security practices, outdated systems and the growing use of IoT medical devices have increased the number of possible entry points for cybercriminals. However, practical solutions that are suitable for healthcare organisations with limited resources are still not well developed in the existing literature.

1.5. THE ROLE OF BACKUP STRATEGIES IN RANSOMWARE RECOVERY

Given the destructive nature of ransomware, effective data strategies are widely recognised as one of the most critical defences available to healthcare organisations. Patel et al. (2024) outlines three main types of backups namely full backups, incremental backups and differential backups and recommends using a layered approach combining multiple backup types along immutable storage and air-gapped systems to ensure that backup data cannot be encrypted or deleted during an attack.

However, evidence from Neprash et al. (2022) suggests that backup strategies in healthcare are failing in practice, finding that only 20.6% of healthcare organisations affected by ransomware were able to restore data from backups and this number decreased over time as attacks became more advanced. Swasey (2020) helps explain this issue by noting that ransomware attackers increasingly target backup files connected to main systems, making it impossible for organisations to recover data without paying the ransom. While Patel et al. (2024) recommends regular testing and updating of backup and recovery plans, there is still a clear gap in the literature. There is limited research on how backup strategies work in developing countries, where limited resources can make it difficult to implement advance solutions. This gap is important and is one that this study aims to address

1.6. COST EFFECTIVE CYBERSECURITY MEASURES FOR HEALTHCARE ORGANISATIONS

The financial challenges faced by healthcare organisations makes it important to identify affordable security measures. Swasey (2020) argues that even small increases in cybersecurity spending, such as allocating an additional 1 to 2% of the IT budget allocation, could significantly reduce the risk of ransomware attacks, recommending practical low-cost measures including employee phishing awareness training, regular software patching and robust backup strategies. Mahmoud (2024) supports this by demonstrating that organisations implementing the ISO 27001 information security management standard, including healthcare institutions such as the NHS in the UK, significantly strengthened their cybersecurity posture through structured risk assessment, patch management and access control frameworks. While Mahmoud (2024) supports a full standard based approach, Swasey (2020) argues that gradually implementing affordable and

practical measures may be more suitable for organisations with limited resources. Despite these recommendations, a significant gap remains in the literature regarding which specific combinations of cybersecurity measures deliver the best outcomes relative to their cost in healthcare settings, leaving organisations without clear evidence based guidance on how to prioritise their cybersecurity spending effectively (Patel, et al., 2024).

1.7. GAPS IDENTIFIED IN THE LITERATURE

Despite the growing body of research on ransomware, several critical gaps remain. The majority of existing studies are conducted in developed countries, particularly the United States and United Kingdom, leaving a significant gap in knowledge regarding ransomware vulnerabilities in developing countries like South Africa (Neprash, et al., 2022). While Singh and Singh (2022) and Chuma (2025) address this gap locally, their studies remain limited in scope and do not examine the combined effect of outdated cybersecurity practices, staff awareness gaps and technical vulnerabilities on ransomware risks. Most studies examine technical vulnerabilities and human factors in isolation rather than as interconnected contributors to ransomware vulnerabilities (Kioskli, et al., 2021). This study addresses these gaps by investigating how outdated cybersecurity practices, technical weaknesses and inadequate staff training collectively contribute to ransomware vulnerabilities in healthcare institutions, while identifying practical and cost-effective measures that institutions can implement.

Research Methodology

1.1. RESEARCH APPROACH AND DESIGN

This study adopts a mixed methods research approach, integrating quantitative and qualitative elements to produce a more complete understanding of the research problem. Research Question 1 is addressed through qualitative methods to explore reasons as to why cybersecurity practices fail while Research Question 2 and 3 are addressed through quantitative methods to analyse the effectiveness of backup strategies and cost implications of security measures. The two components will be integrated using a convergent parallel design, where both types of data are collected and analysed at the same time and then combined during the interpretation stage. The overall design is

both descriptive and explanatory, documenting the current state of cybersecurity practices in healthcare while explaining the underlying causes of ransomware attacks

1.2. TARGET POPULATION AND DATA SOURCES

The target population encompasses healthcare IT professionals, hospital managers and administrators and cybersecurity professionals working within healthcare institutions. However, as this study uses document analysis, no participants will be directly recruited. Instead, data will be drawn from a purposively selected body of peer-reviewed journal articles, academic conference papers, institutional reports, cybersecurity industry reports, published ransomware case studies and publicly available breach databases such as the United States Department of Health and Human Services Office for Civil Rights Data Breach Portal. Sources are selected based on relevance, credibility and quality, limited between 2020 and 2026. While the study draws primarily on international literature, sources addressing both developed and developing country healthcare contexts will be considered to ensure a broader and representative evidence-base.

1.3. DATA COLLECTION METHOD

The main data collection method used in this study is systematic document analysis, a well-established qualitative research technique that involves the systematic examination and interpretation of existing documents to extract relevant information and answer research questions. Data collection will be conducted by searching academic databases including Google Scholar and Research Rabbit using predefined search terms related to ransomware, healthcare cybersecurity, backup strategies and cost-effective security measures. Search results will be screened against inclusion and exclusion criteria to ensure that only relevant, credible and high-quality sources are selected for analysis.

Non-academic documents will be collected from credible institutional sources, including government health departments, cybersecurity agencies, and reputable organisations. These sources will provide additional data in the form of policy reports, breach records and real-world case studies, which are essential for understanding the practical implications of ransomware attacks in healthcare environments

1.4. DATA ANALYSIS STRATEGY

The data collected in this study is analysed using both thematic and comparative analysis. Thematic analysis is applied to qualitative data to identify recurring patterns, key issues related to cybersecurity practices, themes, technical vulnerabilities and staff awareness within healthcare institutions. This approach allows for a deeper understanding of the underlying causes of ransomware vulnerability. In addition, comparative analysis will be used to examine quantitative data, particularly in relation to the effectiveness of backup strategies. This involves comparing outcomes such as recovery time and extent of data loss across different cases. The findings from both qualitative and quantitative analyses will then be integrated to provide a comprehensive evaluation of the problem and to identify the most effective and cost-effective cybersecurity measures.

1.5. VALIDITY, RELIABILITY AND FEASIBILITY

The validity of this study will be ensured by using only credible and verified sources, with clearly documented selection criteria to maintain reliability and consistency. For the qualitative component, trustworthiness will be established using for key criteria. Credibility will be ensured through methodological triangulation, cross-referencing findings from multiple document types. Reliability will be maintained by applying a consistent and systematic approach to data collection and analysis, including the use of predefined keywords, selection criteria and analytical methods. The study is also considered highly feasible as it relies on publicly available secondary data. This eliminates the need for participant recruitment and reduces both time and financial constraints, making it practical complete within the available timeframe.

1.6. ETHICAL CONSIDERATIONS

Although this study does not involve direct data collection from human participants, the ethical risks associated with the research are minimal. However, ethical principles will be strictly followed throughout the study. All sources will be cited in Harvard referencing style, ensuring that the contributions of other researchers are acknowledged and academic integrity is maintained. All sources will be accurately represented without selective reporting or distortion of evidence. Where conflicting findings exist, both perspectives will be acknowledged and discussed objectively. All

written content will be original and produced by the researcher with the ideas and findings from other authors clearly attributed through appropriate in-text citations. Any case studies that contain sensitive information about the healthcare organisations or individuals affected by ransomware attacks, will be handled with discretion and used for solely for academic analysis within the scope of this study. All research materials will be stored securely and accessed only by the researcher and retained in accordance with the institution's data retention policies upon completion of this study.

Expected outcomes

1.1. TYPE OF RESULTS EXPECTED

In relation to the first objective, the study expects to identify a range of critical technical vulnerabilities that consistently expose healthcare organisations to ransomware attacks. Based on evidence from the existing literature, it is anticipated that outdated systems running outdated and unsupported operating systems, unpatched software, weak authentication mechanisms and poor network segmentation will emerge as the most prevalent technical vulnerabilities. It is further expected that human factors, particularly insufficient cybersecurity training and low staff awareness, will be identified as significant contributors to ransomware susceptibility, operating alongside and amplifying the impact of technical weaknesses.

Secondly, the study expects to find that while backup strategies in healthcare is highly recommended, they are often poorly implemented in practice. Many healthcare organisations, especially those operating under resource constraints, lack the technical capacity and financial resources to implement advanced backup solutions such as air gapped and immutable storage systems effectively. Furthermore, effective ransomware recovery is expected to depend on key practices such as regular backup testing, the physical separation of backup systems from primary networks and the existence of well-documented and practised recovery procedures.

Thirdly, the study is expected to identify a set of cost-effective cybersecurity measures suitable for resource-limited healthcare environments. Measures such as regular software patching, employee awareness training, multi-factor authentication and basic backup strategies are anticipated to provide substantial protection at relatively low cost. Additionally, a phased approach to

cybersecurity implementation, prioritising high impact and low-cost interventions, is expected to be more feasible and effective than attempting comprehensive security framework in a single phase.

1.2. HOW FINDINGS WILL ADDRESS THE RESEARCH QUESTIONS

The expected findings of this study are directly aligned with the three research questions outlined in the introduction. The first research question, which asks why current cybersecurity practices in healthcare institutions fail to prevent ransomware attacks effectively, will be addressed by the identification of both technical vulnerabilities and human factors that collectively undermine existing security measures. By examining these factors together rather than in isolation, the study will provide a more comprehensive explanation of why healthcare institutions remain vulnerable despite implementing the basic security controls

The second research question, which focuses on evaluating the effectiveness of backup strategies, will be addressed through quantitative analysis of metrics such as recovery time and data loss rate derived from existing incident reports and case studies. This analysis is expected to produce practical benchmarks that healthcare organisations can use to assess and improve their own backup strategies.

The third research question, which examines the cost-effectiveness of cybersecurity measures, will be addressed through comparative analysis across different healthcare sectors. By evaluating measures in terms of cost, feasibility and the impact, the study is expected to produce a prioritised framework to guide cybersecurity investment decisions.

1.3. POTENTIAL IMPLICATIONS

From a theoretical perspective, the study is expected to contribute to the theoretical understanding of ransomware vulnerabilities in healthcare by providing evidence that technical and human factors interact and compound each other in ways that existing literature has not fully explored. The findings are also expected to extend the theoretical knowledge base by providing evidence from developing countries, contributing to a more globally representative understanding of the problem. Practically, this study is expected to provide healthcare professionals and administrators

with actionable, evidence-based guidance on prioritising cybersecurity measures within limited budgets. The identification of cost-effective measures is particularly significant for smaller and under resourced health care institutions that currently lack guidance on how to strengthen their defences affordably.

From an industry perspective, the findings are expected to highlight specific gaps in the availability of affordable, healthcare-specific cybersecurity solutions. This may encourage the development of more accessible tools and frameworks tailored to resource constraints. In terms of policy, the study is expected to provide an evidence-based to support the development of cybersecurity guidelines and standards tailored to healthcare institutions. Within the South African context, the findings may inform national policy development for public healthcare which currently lack comprehensive and enforceable cybersecurity standards.

Conclusion

This research proposal examines how outdated cybersecurity practices contribute to ransomware vulnerabilities in healthcare institutions. It establishes the growing threat of ransomware and identifies key gaps in current security practices, supported by clearly defined research questions and objectives. The literature review highlights critical issues, including technical and human vulnerabilities, challenges in implementing effective backup strategies and the lack of affordable cybersecurity solutions for resource-constrained organisations. A mixed methods approach is proposed to provide a comprehensive analysis of these issues, combining qualitative and quantitative data. The anticipated findings are expected to address the research question while offering both theoretical insights and practical recommendations. Overall, the study emphasises the need for cybersecurity strategies that account for both technical limitations and resources constraints, contributing to improved protection of patient data and healthcare operations

Declaration of Academic Integrity

Module: RSH381

I, Tshiamo Maise, declare that:

1. I understand that plagiarism is the act of using someone else's words, ideas, or work without proper attribution or acknowledgement, presenting them as one's own. It can involve copying and pasting text, paraphrasing without citation, or closely imitating the structure and ideas of another source without giving credit.
2. I understand that my assignment answers must be accurately referenced and cited.
3. This assignment is my own work. I acknowledge that copying someone else's work, or part of it, is wrong and that submitting identical work to others constitutes a form of plagiarism.
4. I have not allowed and will not allow anyone to copy my work.
5. I understand that I can be awarded 0 marks if I have plagiarised and that a Turnitin similarity rating of 30% or higher will result in 0 marks being awarded.
6. I understand that by typing my name below this text I am acknowledging this as my declaration of honesty and integrity as they relate to academic work.

Name: Tshiamo Maise

Student No: 602178

References

- Aldosari, B., 2025. Cybersecurity in Healthcare: New Threat to Patient Safety. *Cureus*, 17(5).
- Al-Qarni, . E. . A., 2023. Cybersecurity in Healthcare: A Review of Recent Attacks and Mitigation Strategies. *International Journal of Advanced Computer Science and Applications(IJACSA)*, 14(5).
- Alshaikh, H., Ramadan, N. & Hefny, H. A., 2020. Ransomware Prevention and Mitigation Techniques. *International Journal of Computer Applications*, 177(40), pp. 31-39.
- Chuma, K. G., 2025. Legacy electronic health record systems as culprit behind cybersecurity risks in public healthcare facilities of South Africa. *Global Security: Health, Science and Policy*, 10(1).
- Hundal, G. . S., Rhodes, D. & Laux, C., 2025. Digital Transformation of Healthcare Enterprises in the Era of Disruptions—A Structured Literature Review. *Sustainability*, 17(13).
- Jones, D. et al., 2022. CYBERSECURITY THREATS IN THE HEALTHCARE INDUSTRY. *Journal of Business and Educational Leadership*, 12(1), pp. 57-67.
- Kapoor, A. et al., 2021. Ransomware Detection, Avoidance, and Mitigation Scheme: A Review and Future Directions. *Sustainability*, 14(1).
- Kioskli, K., Fotis, T. & Mouratidis, H., 2021. *The landscape of cybersecurity vulnerabilities and challenges in healthcare: Security standards and paradigm shift recommendations*. s.l., s.n., pp. 1-9.
- Mahmoud, K., 2024. WannaCry Impact And How Adoption Of Cyber Crime Measures Such AS. *Fakultet Za Inženjerski Menadžment*.
- Malenfant, T., 2021. *Impact of Ransomware Attacks on Healthcare*, s.l.: Utica College.
- McGlave, C. C. et al., 2023. Characteristics of short-term acute care hospitals that experienced a ransomware attack from 2016 to 2021. *Health Affairs Scholar*, 1(3).
- Mohammed, N., Faisal, A. & B., 2025. Ransomware in Healthcare: Reducing Threats to Patient Care. *Journal of Cognitive Computing and Cybernetic Innovations*, 1(2).

Neprash, H. T., McGlave, C. C. & Cross, D. A., 2022. Trends in Ransomware Attacks on US Hospitals, Clinics, and Other Health Care Delivery Organizations, 2016-2021. *JAMA Health Forum*, 3(12), pp. 50-57.

Patel, S., Bhadouria, A., Dodiya, K. R. & Patel, A., 2024. Evaluating Modern Ransomware and Effective Data Backup and Recovery Solutions. *IJSART*, 10(9).

Pavlova, P. & Albert, C. D., 2025. Defending Health Security: Securing Healthcare Infrastructure against Ransomware. *JSTOR*, 10(2), pp. 41-68.

Shiraz, G., 2024. *Cybersecurity in Healthcare IT: Advanced Threat Mitigation Strategies for Digital Transformation*. s.l.:s.n.

Singh, I. & Singh, Y., 2022. CYBER-SECURITY KNOWLEDGE AND PRACTICE OF NURSES IN PRIVATE HOSPITALS IN NORTHERN DURBAN, KWAZULU-NATAL. *Journal of Theoretical and Applied Information Technology*, 100(1), pp. 246-267.

Sumner, P. & Keenan, R., 2022. Ransomware Attacks on Healthcare Providers—What You Need to Know. *Journal of Health Care Compliance*, pp. 11-16.

Swasey, K., 2020. *Insufficient Healthcare Cybersecurity Invites Ransomware Attacks and Sale of PHI on the Dark Web*, s.l.: s.n.

Technology, I. f. S. a., 2021. *Combating Ransomware*, s.l.: s.n.